



Parametric Open Source Games

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Introduction

- In open-source games, agents may condition their behavior on another agent's internal description, rather than only on external actions.
- This can qualitatively change strategic incentives, producing outcomes that differ from the equilibria of the underlying normal-form game.
- This is relevant for safety and alignment, where agents may reason about, inspect, or verify properties of other agents' internal mechanisms.
- Existing models are usually symbolic and less aligned with modern learning systems, which are parameterized by high-dimensional vectors.

Research Question

Can we formulate a continuous analogue of program equilibria in which agents choose parameter vectors and their behavior may depend on other players' parameters?

Parametric Open Source Games

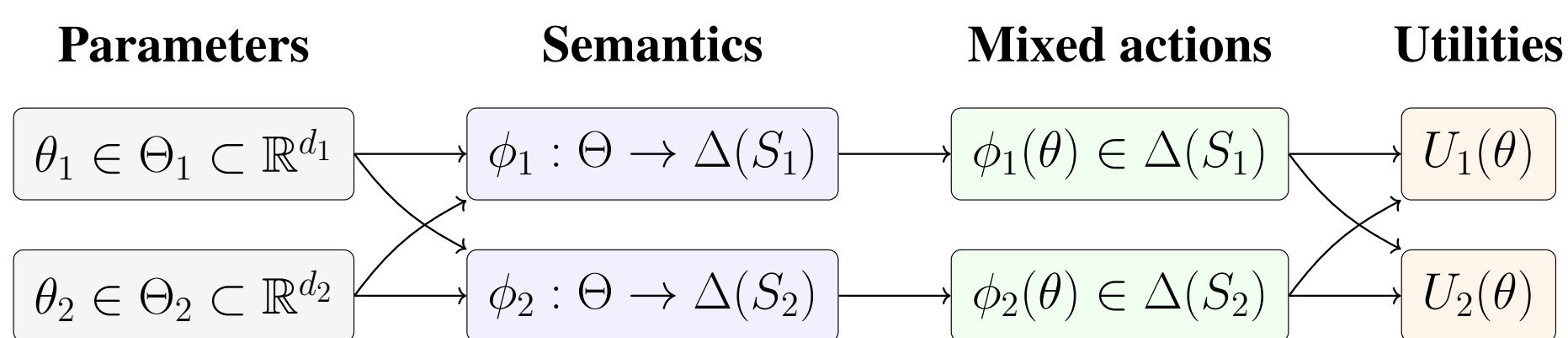


Figure 1: A two-player parametric open-source game. Each player chooses a parameter vector $\theta_i \in \Theta_i$. The semantics map ϕ_i converts the full parameter profile $\theta \in \Theta$ into a mixed action in $\Delta(S_i)$, and the resulting mixed-action profile induces utilities $U_i(\theta)$ in the underlying finite game.

Equilibrium Notion

A profile $\theta^* \in \Theta$ is a parametric program Nash equilibrium if, for every player i ,

$$U_i(\theta_i^*, \theta_{-i}^*) \geq U_i(\theta_i, \theta_{-i}^*) \quad \forall \theta_i \in \Theta_i.$$

If each Θ_i is compact and convex and each ϕ_i is continuous, then a PPNE exists.

- For two-action games with $S_i = \{C, D\}$ and $\theta_i \in [-B, B]$, we define

$$p_i(\theta) = \phi_i(\theta)(C) = \sigma(\theta_i + \gamma\theta_{-i}), \quad \sigma(x) = \frac{1}{1 + e^{-x}}.$$

- The parameter $\gamma \geq 0$ controls how strongly player i 's behavior depends on the opponent's parameter. The case $\gamma = 0$ recovers the closed-source parameterization.
- We optimize the induced parametric game by selfish projected gradient ascent with

$$\theta_i^{t+1} = \Pi_{[-B, B]} \left(\theta_i^t + \eta \frac{\partial U_i(\theta^t)}{\partial \theta_i} \right).$$

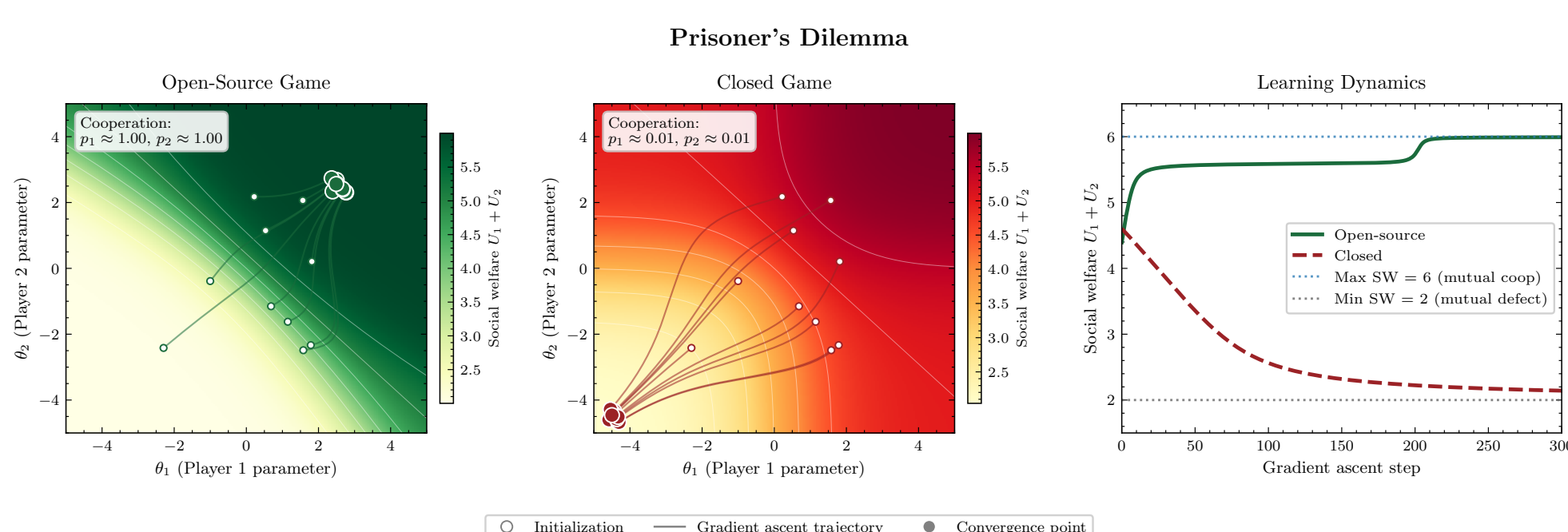


Figure 2: Open-source versus closed-source sigmoid semantics in the Prisoner's Dilemma. With sufficient open-source coupling, selfish gradient ascent moves toward the cooperative boundary. In the closed-source case, the same update moves toward defection.

Nash Equilibria and Basin of Attraction

Phase Transition

For a symmetric 2×2 game with payoffs R, S, T , and P , near $\theta = (0, 0)$, cooperation becomes locally attractive under selfish gradient ascent once

$$\gamma > \gamma^*, \quad \gamma^* = \frac{T + P - R - S}{R + T - P - S}.$$

If $\gamma < \gamma^*$, the local basin instead points toward lower cooperation.

- The phase transition result is a local property.
- It does not by itself show that the boundary point is stable against unilateral deviations.
- The same learning trajectory can suggest cooperation, but equilibrium requires robustness to unilateral parameter deviations.

Equilibrium Verification

The point (B, B) is a PPNE, if

$$\frac{d}{d\theta_i} U_i(\theta_i, B) \geq 0 \quad \forall \theta_i \in (-B, B), \quad i \in \{1, 2\}.$$

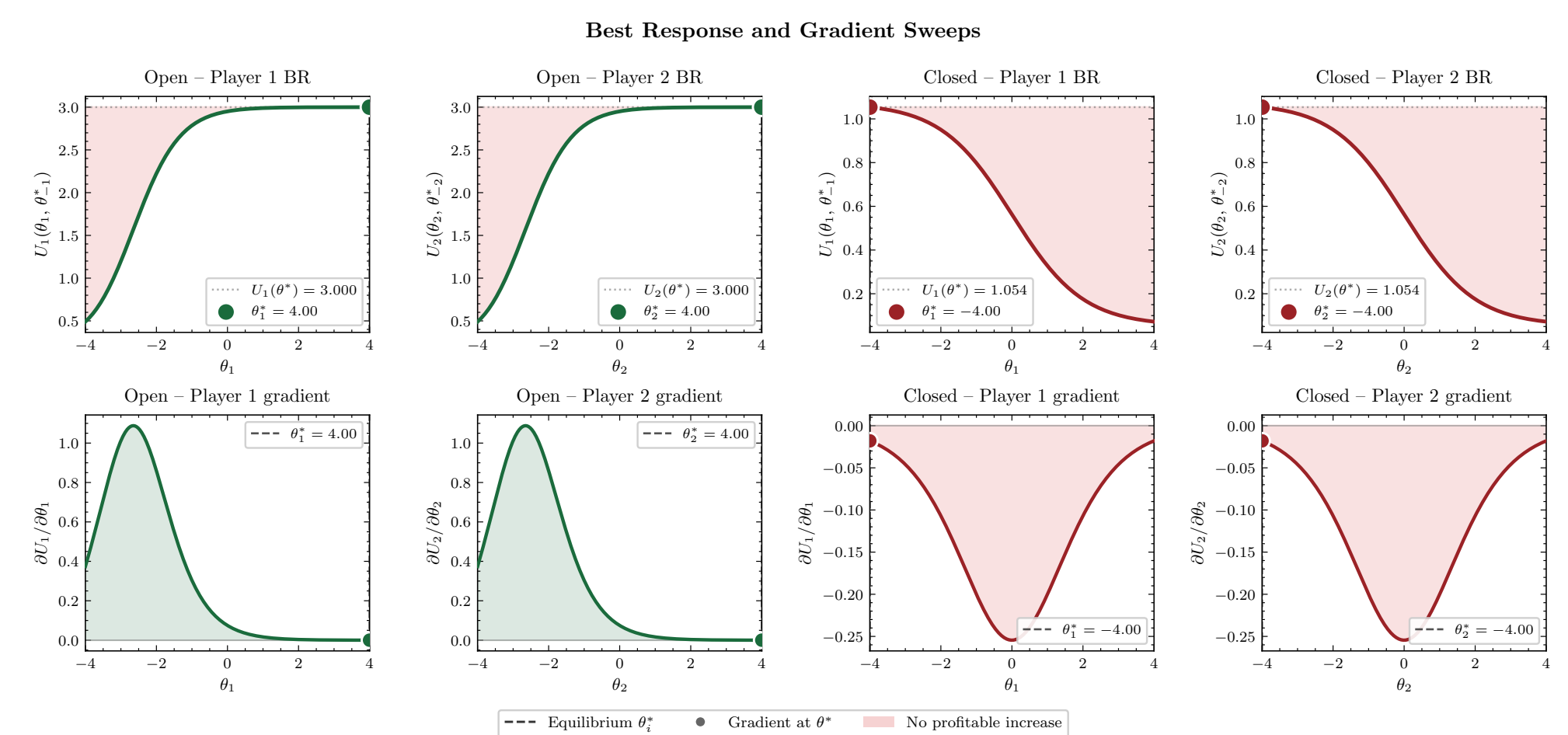


Figure 3: Boundary PPNE verification in the Prisoner's Dilemma. For each candidate equilibrium, the opponent is fixed at its boundary value, and the remaining player's payoff is evaluated over $\theta_i \in [-B, B]$. Open-source semantics maximizes payoff at the cooperative boundary, while closed-source semantics maximizes payoff at the defective boundary.

Neural Semantics and Extensions

- To move beyond linear sigmoid coupling, we introduce richer neural semantics of the form

$$p_i(\theta) = \sigma \left(\alpha\theta_i + \beta\theta_{-i} + h(\theta_i^2, \theta_{-i}^2, \theta_i\theta_{-i}; W) \right),$$

where $h(x; W) = W_2^\top \tanh(W_1^\top x + b_1)$ and $\alpha, \beta \in \mathbb{R}$ are all learnable.

- The neural network uses only quadratic features, so its first-order derivative vanishes at $\theta = (0, 0)$.
- The phase transition result extends as $\frac{\beta}{\alpha} > \gamma^*$, and we verify the model across Stag Hunt, Chicken, Prisoner's Dilemma, and Harsh Prisoner's Dilemma.

Main Message

Open-source game-theoretic dependence need not be symbolic. When encoded by a continuous semantics map, parameter-level transparency can reshape learning dynamics and equilibrium structure. We view this as a starting point for studying partial and weaker transparency, larger populations, and sequential games.